DEMETER: A Blockchain-Based Platform for Tracking the History of Shared Agricultural Data Files

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AGRICULTURAL & BIOLOGICAL ENGINEERING

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Motivation & Questions

- Agricultural data have significant value for farm decisions, environmental assessment, and regulatory compliance, but sharing data across independent organizations remains rare because no organization can confirm where an outside file came from.
- A central database = a single platform that collects and stores all data from participating organizations. This would solve the problem but is impractical at scale. No single organization has the infrastructure to collect, standardize, and maintain all agricultural data.
- Sending files directly lets each organization keep control, but file history information is lost when files move across independent systems. An infrastructure that records file history is needed to enable trusted data sharing across agricultural organizations.

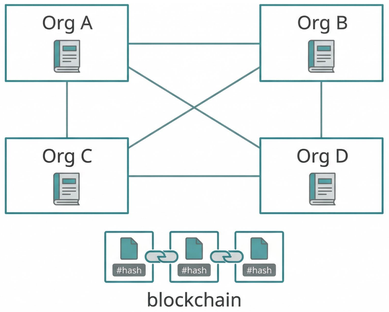
Approach:

1. Identifying what file history information is lost when files move across separate organizations (cross-sectoral review),
2. ...and designing an infrastructure that records file history on a shared blockchain while files remain on each organization's local system.

Questions: How can a receiving organization confirm the origin of a file it did not create? After that file is changed or combined, how can the full history still be traced?

Background: Why a blockchain for sharing files

- Blockchain = a network where all organizations maintain & verify the same records. No need to collect files in one place.
- Records are cryptographically linked. Any modification to past entries is detectable by all organizations.
- File history becomes verifiable & traceable across organizations.



Cross-sectoral Review

- Origin = where a file came from. Who created it and when.
- Transformation = how a file was changed. A new version linked to the earlier file.
- Integration = how files were combined. A combined output linked to its source files.

Sector	Origin	Transformation	Integration
Agriculture	Partial	None	None
Healthcare	Implemented	Partial	Partial
Finance	Partial	Implemented	Implemented

Architecture

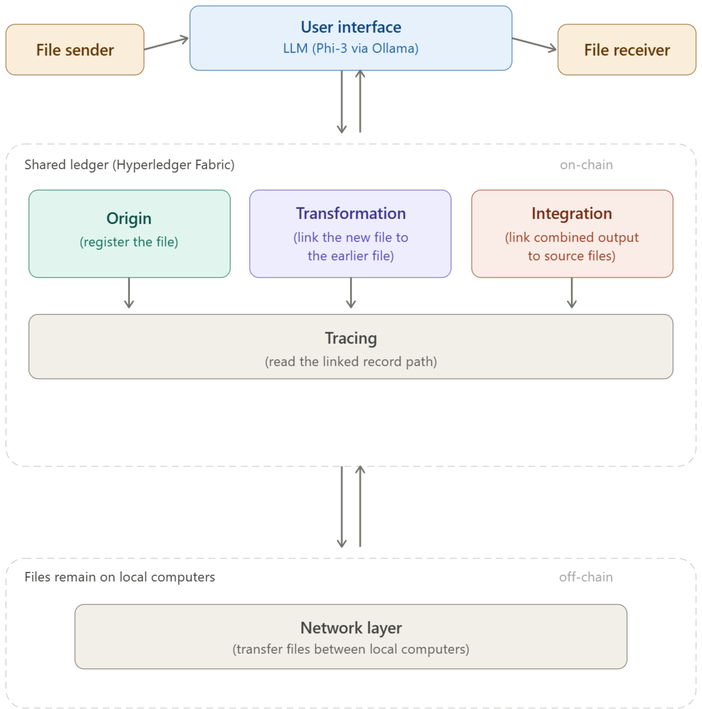
DEMETER keeps file records on a shared blockchain while files remain on local computers.

On-chain (recorded on the blockchain):

- Origin registration → Register the file.
- Transformation declaration → Link the new file to the earlier file.
- Integration composition → Link the combined output to the source files.
- Tracing → Read the linked record path.

Off-chain (outside the blockchain):

- Network layer → Transfer files between local computers.
- Files remain on each organization's local system.



Implementation

- Network: Hyperledger Fabric v2.5, 4 organizations, 1 peer per organization, Raft ordering with 3 ordering nodes.
- Chaincode: Go chaincode per layer, SHA-256 file fingerprinting, X.509 identity model.
- Language interface: Phi-3 via Ollama, local inference only. Seven supported operations: register, verify, transfer, declare transformation, inspect lineage, declare integration, & decompose output.

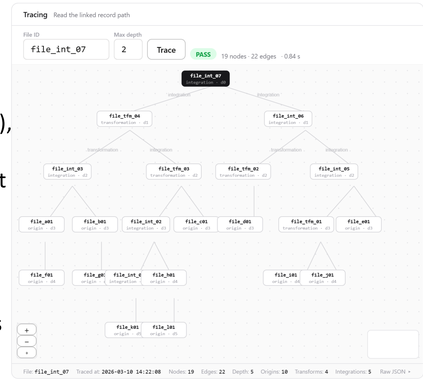
Evaluation & Results

All requirements confirmed across platforms under functional and failure tests.

Claim	Req.	Description	Result
C1	R1, R3	Origin registration remains inspectable across systems	Pass
C2	R4	Transformation lineage inspectable by recipient	Pass
C3	R5	Integration decomposable to source-level verification	Pass
C4	R6	Unverified source flagged without hiding the rest of the trace	Pass
C5	R1–R6	NL interface routes all 7 operations with zero false writes	Pass

Results of tracing

- 19 nodes across 5 depths, traced in 0.84 s. The receiver retrieves the full record path in a single query without accessing the source files.
- 10 origins (R1), 4 transformations (R4), 5 integrations (R5). Each record resolves to a registered file fingerprint on the shared ledger.
- One injected unregistered origin correctly flagged without hiding the rest of the trace (R6).
- Queries under 5 s up to 8,000 records before gateway timeout. Beyond this boundary, pagination-based trace is needed.



Conclusions

- All design requirements (R1–R6) confirmed through C1–C5 across four independently operated systems
- Trace remained functional up to 8,000 linked records.
- The code is available at https://github.com/WhoisHOO/DEMETER

Future Direction

- Trace longer file histories by saving summary points at regular intervals along the record path.
- AI agent interface for automated verification workflows (e.g., soil data sharing, food supply chain governance, agricultural machinery integration).

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